

EEG Signal Classification for Epilepsy Detection Using K-Nearest Neighbors Algorithm

CSCE361101 - Digital Signal Processing (Fall 2025)

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Introduction

This project implements an automated epilepsy detection system that classifies EEG signals into three brain states: Rest, Active, and Seizure. The dataset consists of 500 EEG segments, each containing 4096 samples recorded at 173 Hz sampling rate (~23.6 seconds per segment).

The approach involves: (1) preprocessing using a 50 Hz notch filter to remove electrical interference, (2) extracting five feature representations including raw time-domain signals, statistical features, differentiated signals, derivative statistics, and frequency-band features, and (3) classification using K-Nearest Neighbors (KNN) with K values ranging from 1 to 10.

The dataset was split 80-20 for training and testing. Experiments were conducted using window lengths of 5, 10, 15, and 20 seconds to evaluate their impact on classification performance. This automated approach shows promise for real time seizure detection in clinical settings.

Description of the approach used

The project implements a complete EEG classification pipeline to distinguish rest, active, and seizure brain states from 500 prerecorded EEG segments. Each segment (4096 samples at 173 Hz, ~23.6 s) is first preprocessed with a 50 Hz notch filter to remove mains power line interference while preserving the remaining EEG content. The filtered segments are then divided into windows of length 5, 10, 15, and 20 seconds, and each window is treated as an individual sample for feature extraction and classification.

For every window, five different feature representations are computed. In the time domain, the raw filtered samples are used directly, and four statistical features (mean, variance, skewness, kurtosis) are calculated from the raw signal to summarize its amplitude distribution. Temporal dynamics are captured by computing the first-order difference (derivative) of the signal and then extracting the same four statistics from the differentiated signal. In the frequency domain, the signal is decomposed into standard EEG bands (delta, theta, alpha, beta, gamma) using band-pass filtering, and four statistics per band are computed. These representations allow comparison between raw signals, time-domain summaries, derivative-based features, and frequency-based features.

All window-level feature vectors are labeled according to their parent segment's class (0 = rest, 1 = active, 2 = seizure). For each representation and each window length, the data are split into training and test sets using an 80–20 stratified split that preserves class proportions. A k-Nearest Neighbors (KNN) classifier is then applied with the number of neighbors K varied from 1 to 10. For each K , the classifier predicts the labels of the test windows by majority voting among the K closest training windows in feature space, and classification accuracy is computed as the fraction of correctly classified test samples. Accuracy versus K curves are plotted, and the best K and corresponding accuracy are recorded for each feature type and window length. This framework makes it possible to systematically compare (1) raw vs. time-domain statistical features, (2) derivative-based vs. raw features, (3) time-domain vs. frequency-band features, and (4) the effect of window length on classification performance across all methods.

Deliverables

1. Source Code

https://github.com/Yara-5/DSP_Project/tree/main

2. Plot of the first EEG segment before and after applying a notch filter.

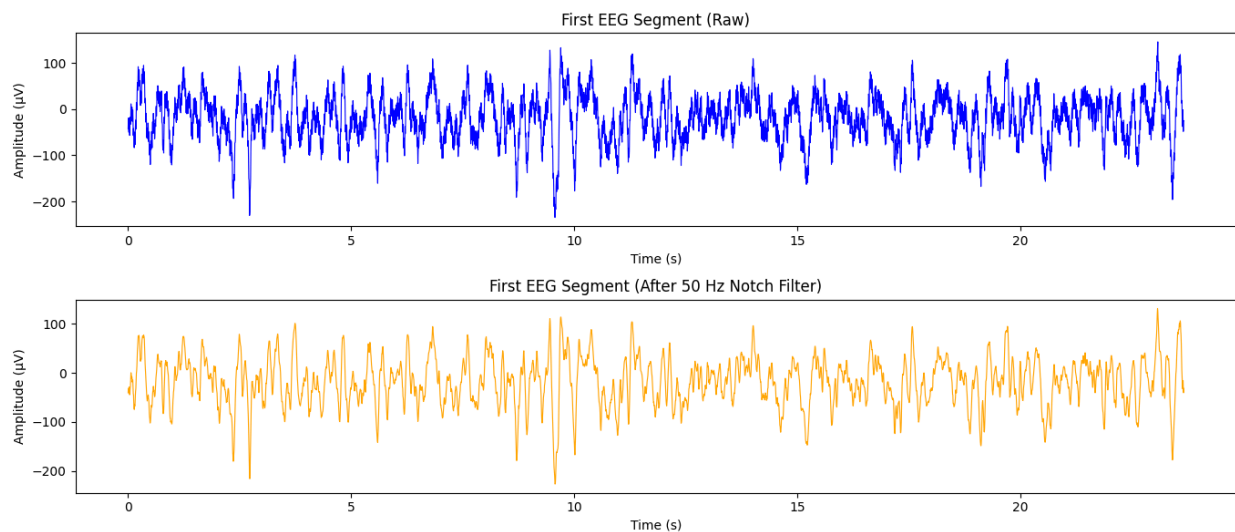


Fig 1. EEG Segment before and after applying 50 HZ Notch Filter

Data Quality Inspection

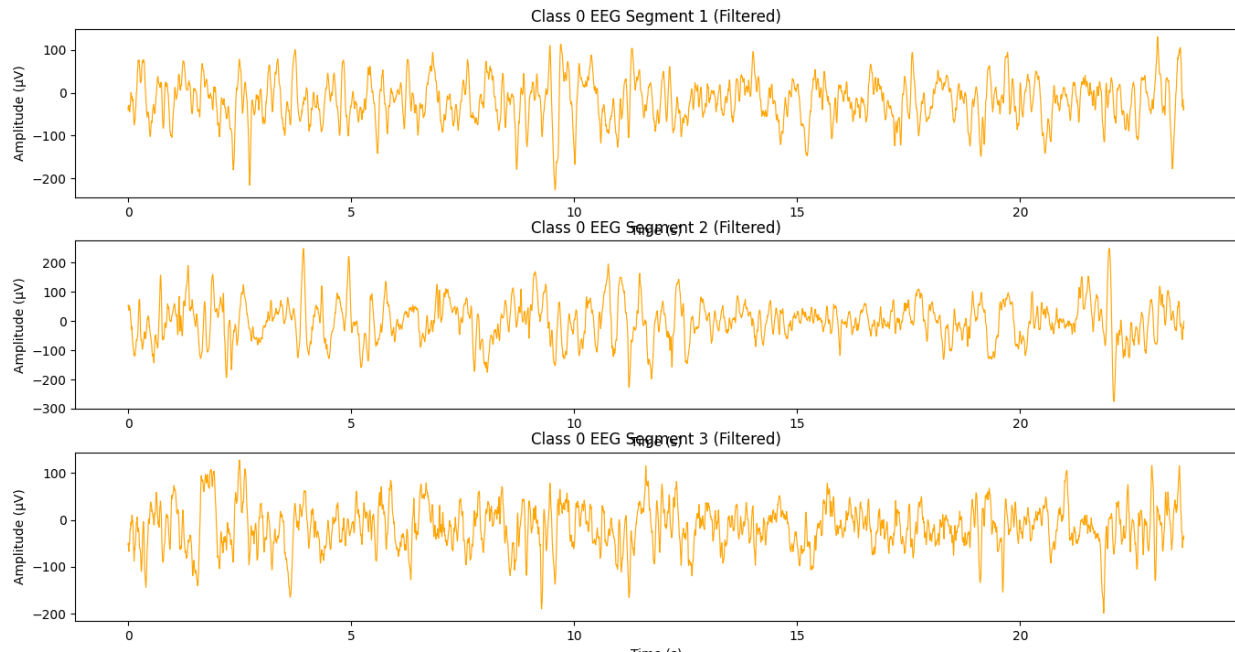


Fig 2. Class 0 EEG Segment.

Class 0: 200 segments

The EEG segments display regular, low-amplitude oscillations ranging between -300 μV and 200 μV . The signals exhibit smooth, rhythmic patterns consistent with alpha wave activity (8-13 Hz) which has characteristics of relaxed and wakeful states with eyes closed. All three segments show consistent minimal artifacts which confirms typical resting state brain activity.

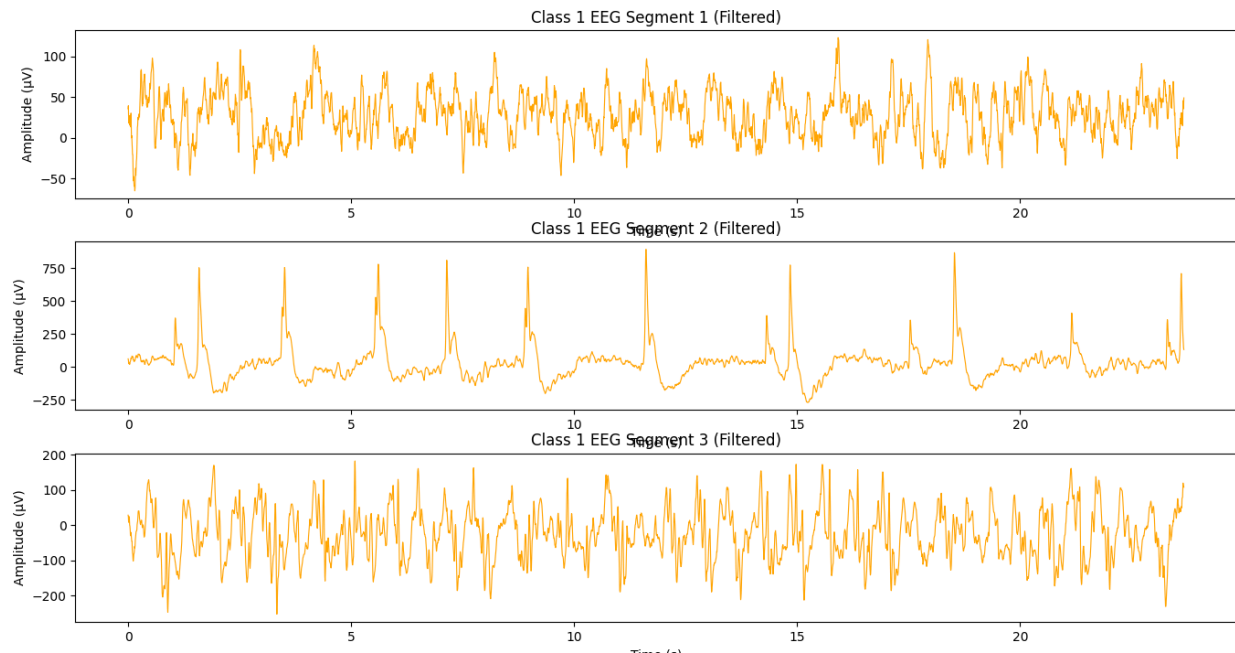


Fig 3. Class 1 EEG Segment.

Class 1: 200 segments

Active state recordings show more variable patterns compared to rest state. Segment 1 displays relatively low amplitudes (± 50 - $100 \mu\text{V}$) similar to rest, while Segments 2 and 3 exhibit intermittent high-amplitude transients (peaks reaching $+750 \mu\text{V}$) likely reflecting muscle artifacts or eye movements during cognitive tasks. The baseline activity between these transients remains similar to rest state, with overall standard deviation of $82.38 \mu\text{V}$.

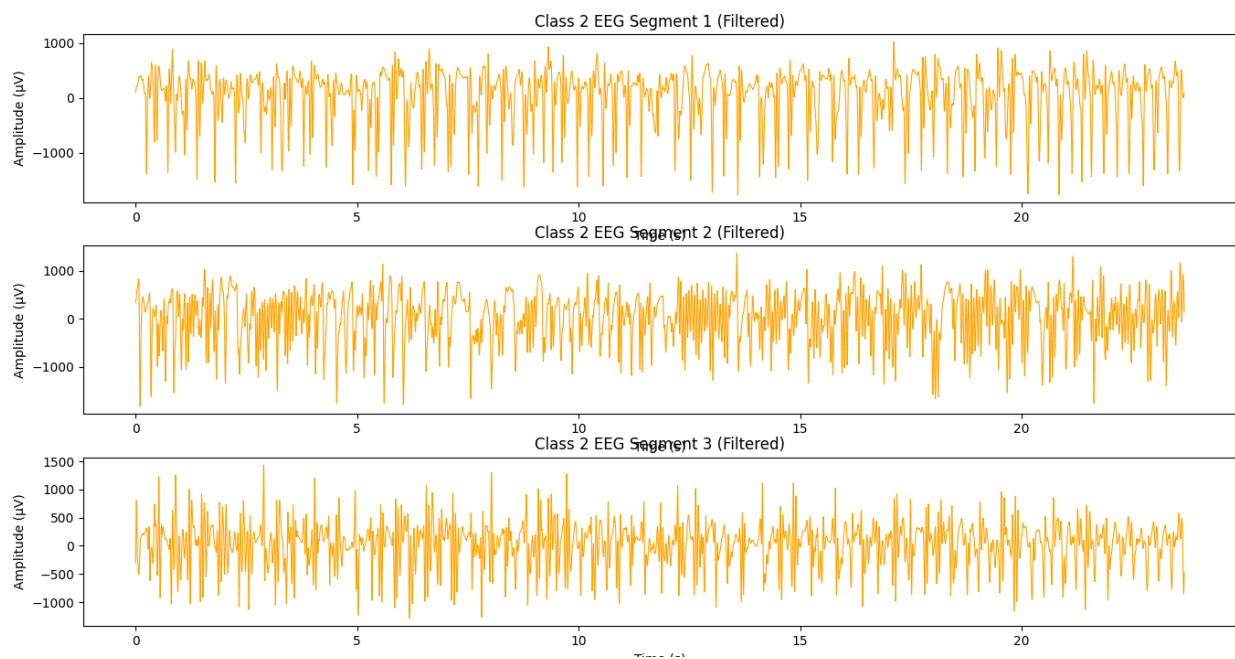


Fig 4. Class 2 EEG Segment.

Class 2: 100 segments

Seizure EEG displays different characteristics with high amplitudes ranging between $\pm 1500 \mu V$ across all three segments approximately 5-10 times larger than normal states. The signals show:

- Persistent high amplitude oscillations throughout the entire 23.6-second duration
- Sharp and irregular bursting activity

• Average spectrum plots:

For each class, plot the average spectrum across all segments belonging to that class.

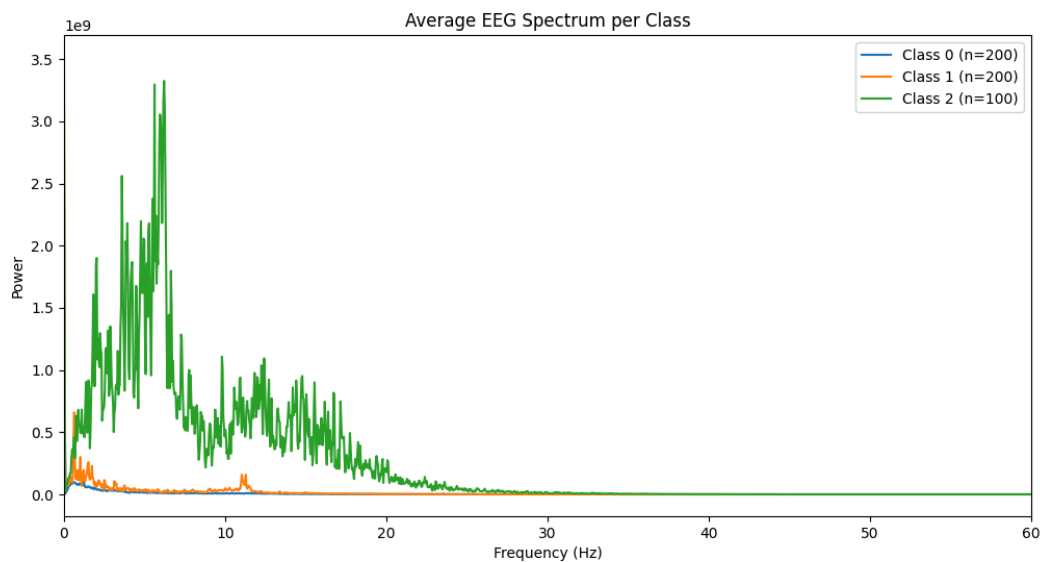


Fig 5. Average EEG Spectrum per Class.

Data amplitude ranges per class:

Class 0: min=-411.00, max=622.38, std=55.99

Class 1: min=-1158.67, max=2066.99, std=82.38

Class 2: min=-1902.87, max=2066.95, std=341.42

Describe how the spectrum was computed (window type, FFT length, averaging method) and comment on the differences among classes. The average power spectrum for each class was computed using the following procedure:

Window Type:

A Hanning (Hann) window was applied to each 4096-sample EEG segment before computing the FFT. The Hanning window reduces spectral leakage by tapering the signal to zero at both ends, minimizing discontinuities that would introduce artificial frequency components.

FFT Length:

The FFT was computed using the full segment length of 4096 samples. This length, being a power of 2 (2^{12}), enables efficient FFT computation and provides a frequency resolution of 0.042 Hz ($173 \text{ Hz} \div 4096 \text{ samples}$).

Averaging Method:

For each segment, the power spectrum was calculated as the squared magnitude of the FFT coefficients ($|FFT|^2$). Individual power spectra were then averaged across all segments belonging to each class using arithmetic mean:

Class 0 (Rest): Average of 200 segments

Class 1 (Active): Average of 200 segments

Class 2 (Seizure): Average of 100 segments

Differences Among Classes

Figure 5 reveals distinct spectral characteristics for each brain state:

Class 0 (Rest State):

Exhibits baseline power levels around 10^7 with relatively flat spectral distribution. The low amplitude reflects normal, desynchronized neural activity during rest, where different brain regions operate independently without coordinated firing patterns.

Class 1 (Active State):

Shows power levels similar to rest state (10^7 to 10^8) with a subtle peak around 12 Hz corresponding to the alpha band. This pattern is consistent with normal cognitive processing where neural activity remains organized but not pathologically synchronized.

Class 2 (Seizure State):

Demonstrates dramatically elevated power approximately 100-300 times higher than normal states with peak power in the 3-7 Hz range (delta-theta bands). The power reaches 3×10^9 , reflecting pathological hypersynchronous activity during epileptic seizures where large populations of neurons fire in abnormal synchronization.

Analysis:

Rest State: Standard deviation = 55.99 μV

Active State: Standard deviation = 82.38 μV

Seizure State: Standard deviation = 341.42

So, seizure signals show power approximately 100 times higher than rest/active states, with seizure power around 10^9 compared to 10^7 for normal brain activity.

KNN Classification and Optimization

Output:

Window Size		Window Size = 5		Window Size = 10		Window Size = 15		Window Size = 20	
Feature Type	K	Accuracy	Best K	Accuracy	Best K	Accuracy	Best K	Accuracy	Best K
Raw EEG	1	0.64	1 or 2	0.665	1 or 2	0.62	1 or 2	0.62	1 or 2
	2	0.64		0.665		0.62		0.62	
	3	0.593		0.56		0.51		0.49	
	4	0.568		0.57		0.48		0.54	
	5	0.54		0.48		0.48		0.46	
	6	0.532		0.49		0.48		0.5	
	7	0.505		0.425		0.48		0.47	
	8	0.512		0.455		0.49		0.53	
	9	0.49		0.39		0.5		0.47	
	10	0.497		0.435		0.49		0.49	
Raw EEG Stats	1	0.608	6	0.615	7	0.54	3 or 4	0.55	3
	2	0.608		0.615		0.54		0.55	
	3	0.61		0.68		0.65		0.65	
	4	0.618		0.69		0.65		0.64	
	5	0.623		0.675		0.62		0.63	
	6	0.642		0.695		0.61		0.62	
	7	0.608		0.715		0.6		0.6	
	8	0.625		0.695		0.57		0.58	
	9	0.598		0.71		0.55		0.6	
	10	0.613		0.705		0.58		0.58	
Derivative signal	1	0.657	1 or 2	0.635	1 or 2	0.54	1 or 2	0.54	1 or 2
	2	0.657		0.635		0.54		0.54	
	3	0.613		0.535		0.52		0.51	
	4	0.613		0.545		0.53		0.52	
	5	0.532		0.495		0.51		0.48	
	6	0.547		0.505		0.52		0.49	
	7	0.502		0.455		0.5		0.48	
	8	0.502		0.47		0.51		0.48	

	9	0.485		0.435		0.5		0.47	
	10	0.492		0.455		0.49		0.48	
Derivative stats	1	0.677	5 or 6	0.64	9	0.6	7	0.63	8 or 10
	2	0.677		0.64		0.6		0.63	
	3	0.68		0.665		0.61		0.65	
	4	0.682		0.685		0.64		0.64	
	5	0.713		0.7		0.65		0.67	
	6	0.713		0.665		0.63		0.66	
	7	0.698		0.685		0.66		0.64	
	8	0.71		0.68		0.65		0.68	
	9	0.69		0.71		0.65		0.66	
	10	0.7		0.7		0.64		0.68	
Frequency-band features	1	0.823	4 or 10	0.83	3 or 4	0.76	3	0.8	1 or 2 or 10
	2	0.823		0.83		0.76		0.8	
	3	0.823		0.84		0.83		0.78	
	4	0.83		0.84		0.82		0.78	
	5	0.818		0.82		0.81		0.79	
	6	0.823		0.805		0.78		0.79	
	7	0.825		0.82		0.77		0.79	
	8	0.823		0.8		0.76		0.79	
	9	0.81		0.815		0.77		0.78	
	10	0.83		0.795		0.78		0.8	

Analysis:

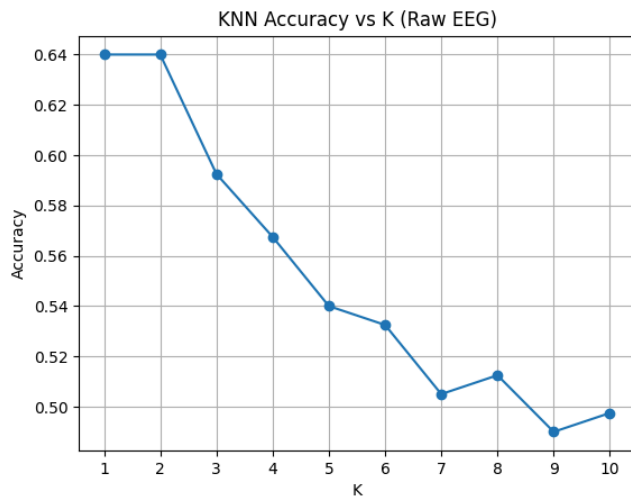
For all experiments, we applied an 80–20 stratified train/test split, ensuring that the class proportions (200, 200, 100) were maintained in both sets.

Each EEG segment was first filtered, windowed, and features extracted. Then the windows were split into train and test sets to avoid data leakage.

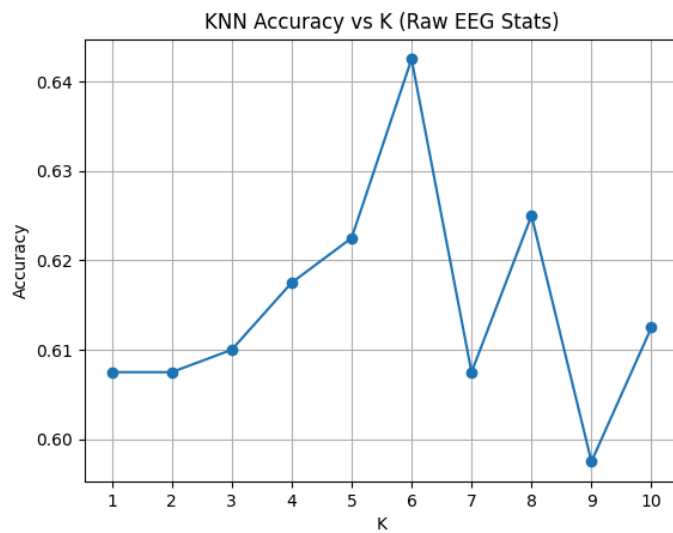
Classification was performed using a KNN classifier with K ranging from 1 to 10.

We can see from the results that for each type of segment, the relationship between k and the accuracy changed.

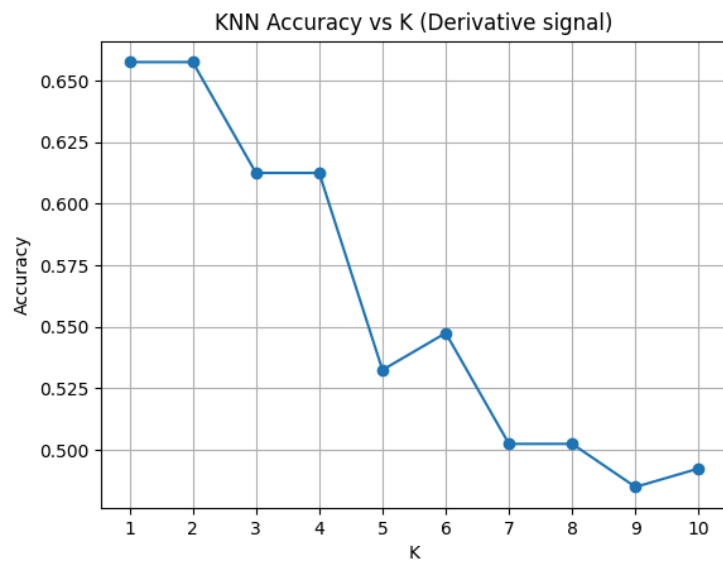
Plots of K vs Accuracy for Window size = 5:



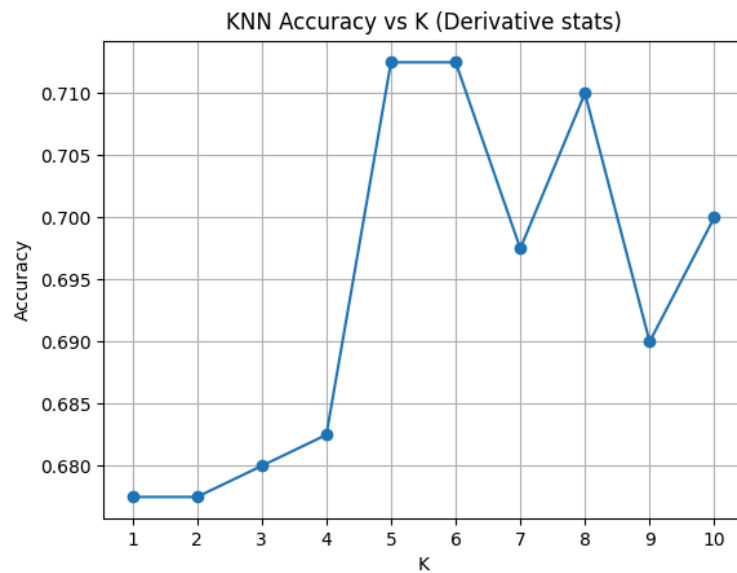
For the raw signal, lower k-values had better accuracies.



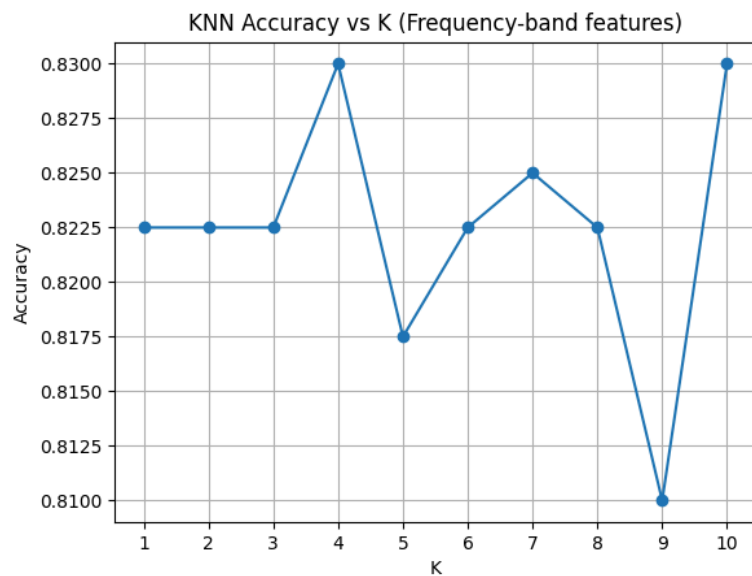
For the raw EEG statistics plot, better accuracies for medium (to low if we consider all window sizes) values of k like k = 6.



For the derivative signal, small k values like 1 and 2 provided the best accuracy results.



As for when applying the KNN on statistics obtained from the derivative signal, medium to high values of k provided better accuracy.



For the frequency bands signal, the values of k did not show any specific correlation with the accuracy.

Time-domain statistical features:

Features used: mean, standard deviation, min, max, range, skewness, kurtosis (all computed per window).

Reason: These features summarize the distribution of EEG samples within a window, reducing noise and dimensionality compared to raw samples.

Performance:

Raw EEG Stats generally improved accuracy relative to raw EEG.

Example: for window = 5 s, raw EEG accuracy was in the range 49%-64%, while EEG Stats had accuracy = 60–64%.

Interpretation: Statistical features help by capturing the general shape and variability of the signal rather than individual noisy points.

Differentiated-signal features:

Features used:

First-order derivative (temporal difference) of the EEG window

Statistical features of the derivative: mean, std, min, max, range, skewness, kurtosis

Reason for choice:

Derivative emphasizes changes and transitions in the signal, highlighting dynamics that may be informative for classification.

Statistics on the derivative summarize these dynamics in a compact way.

Performance:

Raw derivative signal slightly outperformed raw EEG for small K, but derivative statistics were even better, often outperforming raw and raw-stats features.

Example: window = 5 s, derivative stats peak accuracy = 71.3% at K=5, compared to raw EEG which peaked at 64%.

Interpretation: Differentiated statistics capture temporal trends, improving separability of EEG classes.

Window length comparison:

Looking back at the results table, we observe that regardless of the optimal k values, the overall classification accuracy tends to be higher for **shorter window lengths**. This trend is particularly clear for the Raw EEG Statistical Features and Derivative Signal Statistical Features representations.

For some representations, such as Raw EEG and the Derivative Signal, the optimal k remained consistent across all window sizes, suggesting that the classifier's behavior was relatively stable. In contrast, other representations exhibited a wider variation in their optimal k values. The Frequency-band features showed the most variability, with optimal k values appearing to change unpredictably across window sizes.